

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12386441
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Lee; Kah Yong et al.

Wireless lighting effect configuration data transmission system

Abstract

A method, a computer readable medium, and an apparatus for wireless communication are provided. The apparatus may receive, via a radio frequency channel, a first data from a first human interface device at a first time slot of a frame. The apparatus may transmit the first data to a computing device. The apparatus may receive, via the radio frequency channel, a second data from a second human interface device at a second time slot of the frame. The apparatus may transmit the second data to the computing device. The apparatus may transmit, via the radio frequency channel at a third time slot of the frame, at least one of a first acknowledgment to the first data, a first lighting effect configuration data for the first human interface device, a second acknowledgment to the second data, or a second lighting effect configuration data for the second human interface device.

Inventors: Lee; Kah Yong (Singapore, SG), Chan; Chee Oei (Singapore, SG), Dai; Gui Mei (Singapore, SG)

Applicant: Razer (Asia-Pacific) Pte. Ltd. (Singapore, SG)

Family ID: 69181775

Assignee: Razer (Asia-Pacific) Pte. Ltd. (Singapore, SG)

Appl. No.: 18/489766

Filed: October 18, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240045524 A1	Feb. 08, 2024

Related U.S. Application Data

continuation parent-doc US 17865310 20220714 US 11836306 child-doc US 18489766
continuation parent-doc US 17260972 US 11429210 20220830 WO PCT/SG2018/050361

Publication Classification

Int. Cl.: **G06F3/038** (20130101); **G06F3/023** (20060101); **G06F3/0354** (20130101); **G06F13/38** (20060101); **H05B47/19** (20200101)

U.S. Cl.:

CPC **G06F3/0383** (20130101); **G06F3/0231** (20130101); **G06F3/03543** (20130101); **G06F13/385** (20130101); **H05B47/19** (20200101); G06F2203/0337 (20130101); G06F2203/0384 (20130101)

Field of Classification Search

CPC: G06F (3/0383); G06F (3/0231); G06F (3/03543); G06F (13/385); G06F (2203/0337); G06F (2203/0384); G06F (3/038); G06F (3/023); G06F (3/033); H05B (47/19); H05B (45/20)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5881366	12/1998	Bodenmann et al.	N/A	N/A
7015833	12/2005	Bodenmann et al.	N/A	N/A
7024501	12/2005	Wright	N/A	N/A
7480268	12/2008	Jain	N/A	N/A
2005/0090200	12/2004	Karaoguz et al.	N/A	N/A
2005/0243058	12/2004	Morris et al.	N/A	N/A
2005/0243059	12/2004	Morris	345/158	H04W 52/04
2006/0035590	12/2005	Morris et al.	N/A	N/A
2007/0019771	12/2006	Ambuehl et al.	N/A	N/A
2007/0038785	12/2006	Varanda et al.	N/A	N/A
2008/0175207	12/2007	Lee et al.	N/A	N/A
2012/0156993	12/2011	Seo et al.	N/A	N/A
2013/0191554	12/2012	Junge et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
H05119899	12/1992	JP	N/A
H0836446	12/1995	JP	N/A
2002019630	12/2001	WO	N/A
2011130752	12/2010	WO	N/A

OTHER PUBLICATIONS

Chinese Office Action; dated Apr. 17, 2024; Application #: 201880095593.0. cited by applicant
International Search Report and Written Opinion, dated Apr. 9, 2019, for the corresponding International Application No. PCT/SG2018/050361 in 8 pages. cited by applicant
Extended European Search Report mailed Jun. 25, 2021, 11 pages, for the corresponding European Patent Application No. 18928114.0. cited by applicant

Benjamin F. Janzen et al., “Is 60 FPS Better than 30? The Impact of Frame Rate and Latency on Moving Target Selection.” CHI 2014, One of a CHIInd, Toronto, ON, Canada, CHI 14 Extended Abstracts on Human Factors in Computing Systems. 2014. 1477-1482. cited by applicant
Denis Barberena, et al. “Optical Verification of Mouse Event Accuracy.” In 5 pages, 2016. cited by applicant
Alienware AW568 Advanced Gaming Mechanical Keyboard + AW958 Elite 12000 DPI Gaming Mouse, (<https://www.ipohonline.biz/alienware-aw568-advanced-gaming-keyboard-and-elite-gamingmouse-combo>), ipohonline.biz Marketplace Solution Provider in 14 pages; 2018. cited by applicant
Mark Beaumont et al., “The Cross Domain Desktop Compositor: Using hardware-based video compositing for a multi-level secure user interface.” Proceedings of the 32nd Annual Conference on Computer Security Applications. 2016 in 13 pages. cited by applicant

Primary Examiner: Giesy; Adam R.

Attorney, Agent or Firm: Polsinelli PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/865,310 filed Jul. 14, 2022, which is a continuation of U.S. patent application Ser. No. 17/260,972 filed Jan. 15, 2021 which has issued on Aug. 30, 2022 as U.S. Pat. No. 11,429,210, which is a National Phase application under 35 U.S.C. § 371 of PCT/SG2018/050361, filed on Jul. 23, 2018, which published as WO 2020/022950 on Jan. 20, 2020, the contents of each are incorporated herein by reference to their entirety.

TECHNICAL FIELD

(1) Various aspects of this disclosure generally relate to wireless communication systems, and more particularly, to a wireless lighting effect configuration data transmission system.

BACKGROUND

(2) A human interface device (HID) is a type of computer device usually used by humans that takes input from humans and gives output to humans. Common human interface devices may include keyboard, pointing devices (e.g., mouse, trackball, touchpad, pointing stick, light pen), touchscreen, magnetic stripe reader, graphics tablet, joystick, gamepad, analog stick, webcam, fingerprint scanner, etc. A dongle is a small piece of hardware that connects to a computing device to provide the computing device with additional functionality (e.g., wireless connectivity to human interface devices).

(3) A wireless mouse may transmit data via infrared radiation (IR) or radio (e.g., Bluetooth and Wi-Fi). The receiver of a wireless mouse may be a dongle that is connected to the computer through a serial or USB port, or can be built in. A wireless keyboard is a computer keyboard that allows the user to communicate with computers, tablets, or laptops with the help of radio frequency (RF) or infrared radiation technology. A wireless keyboard may be connected using RF technology using a transmitter and a receiver. The radio transmitter may be inside the wireless keyboard. The radio receiver may be a dongle that is plugged into a keyboard port or USB port. Once the receiver and transmitter are plugged in, the computer recognizes the wireless keyboard as if it were connected via a cable.

(4) The wireless mouse and the wireless keyboard may use light-emitting diode (LED) components to present lighting effect. In traditional wireless mouse and wireless keyboard architecture, there may be a dedicated USB dongle associated to each HID device. The wireless mouse and the

wireless keyboard may support wireless Chroma function. That is, the wireless mouse and the wireless keyboard may adjust their lighting effect based on configuration data received from their respective USB dongle. The wireless mouse and the wireless keyboard operate independently from each other.

(5) FIG. 1A is a diagram illustrating an example of a traditional system **100** that includes a wireless mouse **106** and a wireless keyboard **104**. In the example, the wireless mouse **106** is connected to a computing device **102** through a USB dongle **110**, the wireless keyboard **104** is connected to the computing device **102** through a USB dongle **108**. The USB dongle **110** is the receiver of the wireless mouse **106**, and the USB dongle **108** is the receiver of the wireless keyboard **104**.

(6) The wireless mouse **106** and the wireless keyboard **104** may interfere with each other as the RF transmission is not coordinated. The close proximity of the dongles **108** and **110** may create unwanted RF degradation to the systems, which is known as power blocking. In addition, two USB ports of the computing device **102** are occupied as each HID device needs one USB dongle.

SUMMARY

(7) The following presents a simplified summary of one or more aspects in order to provide a basic understanding of such aspects. This summary is not an extensive overview of all contemplated aspects, and is intended to neither identify key or critical elements of all aspects nor delineate the scope of any or all aspects. Its sole purpose is to present some concepts of one or more aspects in a simplified form as a prelude to the more detailed description that is presented later.

(8) In an aspect of the disclosure, a system and a method to transmit lighting effect configuration data to multiple human interface devices (HID) using a single dongle (e.g., a USB dongle) are provided. The system and the method allow two input devices (e.g., a wireless mouse and a wireless keyboard) to communicate with a single dongle on one communication channel. The performance of the two input devices is on-par with wired devices. The wireless system may divide the channel bandwidth into transmission timeslots, and dedicated timeslots may be allocated to input devices for data transmission, and the acknowledgement and lighting effect configuration data may be returned to the input devices in a common slot.

(9) In another aspect of the disclosure, a method, a computer readable medium, and an apparatus for wireless communication are provided. The apparatus may be a dongle. The apparatus may receive, via a radio frequency channel, a first data from a first human interface device at a first time slot of a frame. The apparatus may transmit the first data to a computing device coupled to the apparatus. The apparatus may receive, via the radio frequency channel, a second data from a second human interface device at a second time slot of the frame. The apparatus may transmit the second data to the computing device. The apparatus may transmit, via the radio frequency channel at a third time slot of the frame, at least one of a first acknowledgment to the first data, a first lighting effect configuration data for the first human interface device, a second acknowledgment to the second data, or a second lighting effect configuration data for the second human interface device.

(10) To the accomplishment of the foregoing and related ends, the one or more aspects include the features hereinafter fully described and particularly pointed out in the claims. The following description and the annexed drawings set forth in detail certain illustrative features of the one or more aspects. These features are indicative, however, of but a few of the various ways in which the principles of various aspects may be employed, and this description is intended to include all such aspects and their equivalents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a diagram illustrating an example of a traditional system that includes a wireless

mouse and a wireless keyboard.

(2) FIG. 1B is a diagram illustrating an example of a data transmission system that includes a wireless mouse and a wireless keyboard.

(3) FIG. 2 is a diagram illustrating an example of an architecture for data transmission between a dongle and two human interface devices (e.g., a wireless mouse and a wireless keyboard).

(4) FIG. 3 is a flowchart of a method of operating a dongle at the first time slot in the data transmission system described above in FIG. 2.

(5) FIG. 4 is a flowchart of a method of operating a dongle at the second time slot in the data transmission system described above in FIG. 2.

(6) FIG. 5 is a flowchart of a method of operating a dongle at the third time slot in the data transmission system described above in FIG. 2.

(7) FIG. 6 is a flowchart of a method of operating a wireless mouse at the first time slot in the data transmission system described above in FIG. 2.

(8) FIG. 7 is a flowchart of a method of operating a wireless mouse at the third time slot in the data transmission system described above in FIG. 2.

(9) FIG. 8 is a flowchart of a method of operating a wireless keyboard at the second time slot in the data transmission system described above in FIG. 2.

(10) FIG. 9 is a flowchart of a method of operating a wireless keyboard at the third time slot in the data transmission system described above in FIG. 2.

(11) FIG. 10 is a diagram illustrating an example of data transmissions between a dongle, a mouse, and a keyboard.

(12) FIG. 11 is a flowchart of a method of wireless communication.

(13) FIG. 12 is a conceptual data flow diagram illustrating the data flow between different means/components in an exemplary apparatus.

(14) FIG. 13 is a diagram illustrating an example of a hardware implementation for an apparatus employing a processing system.

DETAILED DESCRIPTION

(15) The detailed description set forth below in connection with the appended drawings is intended as a description of various configurations and is not intended to represent the only configurations in which the concepts described herein may be practiced. The detailed description includes specific details for the purpose of providing a thorough understanding of various concepts. However, it will be apparent to those skilled in the art that these concepts may be practiced without these specific details. In some instances, well known structures and components are shown in block diagram form in order to avoid obscuring such concepts.

(16) Several aspects of a wireless communication system will now be presented with reference to various apparatus and methods. These apparatus and methods will be described in the following detailed description and illustrated in the accompanying drawings by various blocks, components, circuits, processes, algorithms, etc. (collectively referred to as “elements”). These elements may be implemented using electronic hardware, computer software, or any combination thereof. Whether such elements are implemented as hardware or software depends upon the particular application and design constraints imposed on the overall system.

(17) By way of example, an element, or any portion of an element, or any combination of elements may be implemented as a “processing system” that includes one or more processors. Examples of processors include microprocessors, microcontrollers, graphics processing units (GPUs), central processing units (CPUs), application processors, digital signal processors (DSPs), reduced instruction set computing (RISC) processors, systems on a chip (SoC), baseband processors, field programmable gate arrays (FPGAs), programmable logic devices (PLDs), state machines, gated logic, discrete hardware circuits, and other suitable hardware configured to perform the various functionality described throughout this disclosure. One or more processors in the processing system may execute software. Software shall be construed broadly to mean instructions, instruction sets,

code, code segments, program code, programs, subprograms, software components, applications, software applications, software packages, routines, subroutines, objects, executables, threads of execution, procedures, functions, etc., whether referred to as software, firmware, middleware, microcode, hardware description language, or otherwise.

(18) Accordingly, in one or more example embodiments, the functions described may be implemented in hardware, software, or any combination thereof. If implemented in software, the functions may be stored on or encoded as one or more instructions or code on a computer-readable medium. Computer-readable media includes computer storage media. Storage media may be any available media that can be accessed by a computer. By way of example, and not limitation, such computer-readable media may include a random-access memory (RAM), a read-only memory (ROM), an electrically erasable programmable ROM (EEPROM), optical disk storage, magnetic disk storage, other magnetic storage devices, combinations of the aforementioned types of computer-readable media, or any other medium that can be used to store computer executable code in the form of instructions or data structures that can be accessed by a computer.

(19) FIG. 1B is a diagram illustrating an example of a data transmission system **150** that includes a wireless mouse **156** and a wireless keyboard **154**. In the example, both the wireless mouse **156** and the wireless keyboard **154** are connected to a computing device **152** through a shared USB dongle **158**. The USB dongle **158** is the receiver of both the wireless mouse **156** and the wireless keyboard **154**.

(20) The data transmission system **150** makes use of a single USB dongle **158** to communicate with the wireless mouse **156** and the wireless keyboard **154** concurrently. The wireless mouse **156** may upload mouse data that includes movement and button status to the USB dongle **158** at the radio frequency of 1000 Hz. The wireless keyboard **154** may upload keys pressed status to the USB dongle **158** at the radio frequency of 1000 Hz. The USB dongle **158** may download lighting effect configuration data to the wireless mouse **156** and the wireless keyboard **154** concurrently at the frequency of 30 frames per second. Each transmission may be completed in a given time slot.

(21) In some embodiments, a data transmission system may include a USB dongle, which may be able to support two or more human interface devices (e.g., one wireless mouse and one wireless keyboard) concurrently. In some embodiments, the human interface devices may transmit HID data to the USB dongle at the radio frequency of 1000 Hz. In some embodiments, the USB dongle may stream HID lighting effect configuration data (may be referred to as Chroma data) to the human interface devices at the frequency of 30 frames per second. The human interface devices may use the HID lighting effect configuration data to adjust their respective lighting effect.

(22) FIG. 2 is a diagram **200** illustrating an example of an architecture for data transmission between a dongle and two human interface devices (e.g., a wireless mouse and a wireless keyboard). In the example, data transmission is divided into frames of $T \mu s$ (microsecond). Each frame may be sub-divided into three time slots for mouse, keyboard, and Chroma data. The duration of each time slot may be allocated according to the data bandwidth required by the function associated with the time slot.

(23) In some embodiments, at the start of each frame, the dongle may enter receive mode (RX) in order to receive data from mouse and keyboard respectively. The dongle may stay at the receive mode for two time slots (slot 1 and slot 2). The dongle may change to transmit mode (TX) at slot 3 in order to transmit acknowledgment (ACK) and Chroma data to the mouse and/or keyboard.

(24) One of ordinary skill in the art would realize that each frame may be sub-divided into N time slots, N being greater than or equal to three. Among the N time slots, $N-1$ time slots may be used to receive data from $N-1$ human interface devices, which may work concurrently with a computing device via a dongle coupled to the computing device. One time slot may be used to transmit ACK and Chroma data from the dongle to the $N-1$ human interface devices.

(25) FIG. 3 is a flowchart **300** of a method of operating a dongle at the first time slot in the data transmission system described above in FIG. 2. The method may be performed by an apparatus

(e.g., apparatus **1202/1202'** described below with reference to FIG. **12** or **13**). The apparatus may be the dongle.

(26) At **302**, the apparatus may determine whether it is connected to a wireless mouse. If it is determined that the apparatus is connected to a wireless mouse, the apparatus may proceed to **306**. If it is determined that the apparatus is not connected to a wireless mouse, the apparatus may proceed to **304**.

(27) At **306**, the apparatus may wait for mouse data (e.g., data regarding the movement and button status of the wireless mouse) to arrive. At **314**, the apparatus may determine whether the mouse data is received. If the mouse data is received, the apparatus may proceed to **316**. At **316**, the apparatus may set an acknowledgment flag for the mouse data.

(28) At **304**, the apparatus may determine whether a wireless mouse is trying to connect to the apparatus. If it is determined that a wireless mouse is trying to connect, the apparatus may proceed to **310**. If it is determined that a wireless mouse is not trying to connect, the apparatus may proceed to **308**.

(29) At **310**, the apparatus may perform a connection sequence to get connected with the wireless mouse. At **312**, the apparatus is connected to the wireless mouse. At **308**, the apparatus may transmit beacon for connection and synchronization with a wireless mouse.

(30) FIG. **4** is a flowchart **400** of a method of operating a dongle at the second time slot in the data transmission system described above in FIG. **2**. The method may be performed by an apparatus (e.g., apparatus **1202/1202'** described below with reference to FIG. **12** or **13**). The apparatus may be the dongle.

(31) At **402**, the apparatus may determine whether it is connected to a wireless keyboard. If it is determined that the apparatus is connected to a wireless keyboard, the apparatus may proceed to **408**. If it is determined that the apparatus is not connected to a wireless keyboard, the apparatus may proceed to **406**.

(32) At **408**, the apparatus may wait for keyboard data (e.g., data regarding keys pressed status of the wireless keyboard) to arrive. At **418**, the apparatus may determine whether the keyboard data is received. If the keyboard data is received, the apparatus may proceed to **420**. At **420**, the apparatus may set an acknowledgment flag for the keyboard data.

(33) At **406**, the apparatus may determine whether a wireless keyboard is trying to connect to the apparatus. If it is determined that a wireless keyboard is trying to connect, the apparatus may proceed to **412**. If it is determined that a wireless keyboard is not trying to connect, the apparatus may proceed to **410**.

(34) At **412**, the apparatus may perform a connection sequence to get connected with the wireless keyboard. At **416**, the apparatus is connected to the wireless keyboard. At **410**, the apparatus may transmit beacon for connection and synchronization with a wireless keyboard.

(35) FIG. **5** is a flowchart **500** of a method of operating a dongle at the third time slot in the data transmission system described above in FIG. **2**. The method may be performed by an apparatus (e.g., apparatus **1202/1202'** described below with reference to FIG. **12** or **13**). The apparatus may be the dongle.

(36) At **502**, the apparatus may determine whether an acknowledgment flag for mouse data or keyboard data is set. If it is determined that an acknowledgment flag for mouse data or keyboard data is set, the apparatus may proceed to **504**. If it is determined that no acknowledgment flag for mouse data or keyboard data is set, the apparatus may proceed to **506**.

(37) At **506**, the apparatus may set RF to idle at the third time slot. At **504**, the apparatus may determine whether Chroma data for the wireless mouse or wireless keyboard is ready. If the Chroma data for the wireless mouse or wireless keyboard is ready, the apparatus may proceed to **508**. If no Chroma data for the wireless mouse or wireless keyboard is ready, the apparatus may proceed to **510**.

(38) At **508**, the apparatus may send acknowledgement to the mouse data and/or the keyboard data,

as well as Chroma data. At **510**, the apparatus may send only acknowledgement to the mouse data and/or the keyboard data.

(39) FIG. **6** is a flowchart **600** of a method of operating a wireless mouse at the first time slot in the data transmission system described above in FIG. **2**. The method may be performed by the wireless mouse.

(40) At **602**, the mouse may determine whether it is connected to a dongle wirelessly. If it is determined that the mouse is connected to a dongle, the mouse may proceed to **608**. If it is determined that the mouse is not connected to a dongle, the mouse may proceed to **604**.

(41) At **608**, the mouse may determine whether mouse data (e.g., data regarding the movement and button status of the wireless mouse) is ready to send. If mouse data is ready to send, the mouse may proceed to **618**. If mouse data is not ready to send, the mouse may proceed to **620**.

(42) At **618**, the mouse may send mouse data to the dongle. At **622**, the mouse may set a flag for waiting acknowledgment to the mouse data. At **620**, the mouse may set RF to idle at the first time slot.

(43) At **604**, the mouse may determine whether a beacon is received. If a beacon is received, the mouse may proceed to **612**. If a beacon is not received, the mouse may proceed to **610**.

(44) At **612**, the mouse may perform a connection sequence to connect to the dongle. At **614**, the mouse is connected to the dongle. At **610**, the mouse may listen for beacon data.

(45) FIG. **7** is a flowchart **700** of a method of operating a wireless mouse at the third time slot in the data transmission system described above in FIG. **2**. The method may be performed by the wireless mouse.

(46) At **702**, the mouse may determine whether the flag for waiting acknowledgment is set. If the flag for waiting acknowledgment is set, the mouse may proceed to **706**. If the flag for waiting acknowledgment is not set, the mouse may proceed to **704**.

(47) At **706**, the mouse may set RF to receive mode and wait for the acknowledgment data. At **708**, the mouse may determine whether acknowledgment is received. If acknowledgment is received, the mouse may proceed to **710**. At **710**, the mouse may process the received acknowledgment and Chroma data for the mouse. At **704**, the mouse may set RF to idle at the third time slot.

(48) FIG. **8** is a flowchart **800** of a method of operating a wireless keyboard at the second time slot in the data transmission system described above in FIG. **2**. The method may be performed by the wireless keyboard.

(49) At **802**, the keyboard may determine whether it is connected to a dongle wirelessly. If it is determined that the keyboard is connected to a dongle, the keyboard may proceed to **806**. If it is determined that the keyboard is not connected to a dongle, the keyboard may proceed to **804**.

(50) At **806**, the keyboard may determine whether keyboard data (e.g., data regarding keys pressed status of the wireless keyboard) is ready to send. If keyboard data is ready to send, the keyboard may proceed to **816**. If keyboard data is not ready to send, the keyboard may proceed to **820**.

(51) At **816**, the keyboard may send keyboard data to the dongle. At **822**, the keyboard may set a flag for waiting acknowledgment to the keyboard data. At **820**, the keyboard may set RF to idle at the second time slot.

(52) At **804**, the keyboard may determine whether a beacon is received. If a beacon is received, the keyboard may proceed to **812**. If a beacon is not received, the keyboard may proceed to **810**.

(53) At **812**, the keyboard may perform a connection sequence to connect to the dongle. At **814**, the keyboard is connected to the dongle. At **810**, the keyboard may listen for beacon data.

(54) FIG. **9** is a flowchart **900** of a method of operating a wireless keyboard at the third time slot in the data transmission system described above in FIG. **2**. The method may be performed by the wireless keyboard.

(55) At **902**, the keyboard may determine whether the flag for waiting acknowledgment is set. If the flag for waiting acknowledgment is set, the keyboard may proceed to **906**. If the flag for waiting acknowledgment is not set, the keyboard may proceed to **904**.

(56) At **906**, the keyboard may set RF to receive mode and wait for the acknowledgment data. At **908**, the keyboard may determine whether acknowledgment is received. If acknowledgment is received, the keyboard may proceed to **910**. At **910**, the keyboard may process the received acknowledgment and Chroma data for the keyboard. At **904**, the keyboard may set RF to idle at the third time slot.

(57) FIG. **10** is a diagram **1000** illustrating an example of data transmissions between a dongle **1002**, a mouse **1004**, and a keyboard **1006**. The mouse **1004** may be a wireless mouse and the keyboard **1006** may be a wireless keyboard. In some embodiments, the data transmissions in the example may occur in the data transmission system described above with reference to FIG. **2**.

(58) At frame (N-1), both the mouse **1004** and the keyboard **1006** have HID data to transmit to the dongle **1002**. At frame (N-1), slot 1, the mouse **1004** transmits HID data to the dongle **1002**. At frame (N-1), slot 2, the keyboard **1006** transmits HID data to the dongle **1002**.

(59) At frame (N-1), slot 3, the mouse **1004** and the keyboard **1006** change to RX mode. The dongle **1002** transmits ACK to the mouse **1004** and the keyboard **1006**, as well as Chroma data to the mouse **1004**. The mouse **1004** receives ACK, which indicates the success of previous HID transmission. The mouse **1004** also receives the Chroma data to configure the lighting effect of the mouse **1004**. The keyboard **1006** receives ACK, which indicate the success of previous HID transmission.

(60) At frame (N), only the mouse **1004** has HID data to transmit. At frame (N), slot 1, the mouse **1004** transmits HID data to the dongle **1002**. At frame (N), slot 2, no transmission from the keyboard **1006**. The keyboard **1006** is placed in idle mode. At frame (N), slot 3, the mouse **1004** changes to RX mode. The dongle **1002** transmits ACK to the mouse **1004**, as well as Chroma data to the mouse **1004**. The mouse **1004** receives ACK, which indicate the success of previous HID transmission. The mouse **1004** also receives Chroma data to configure the lighting effect of the mouse **1004**.

(61) At frame (N+1), only the keyboard **1006** has HID data to transmit. At frame (N+1), slot 1, no transmission from the mouse **1004**. The mouse **1004** is placed in idle mode. At frame (N+1), slot 2, the keyboard **1006** transmits HID data to the dongle **1002**. At frame (N+1), slot 3, the keyboard **1006** changes to RX mode. The dongle **1002** transmits ACK to the keyboard **1006**, as well as Chroma data to the keyboard **1006**. The keyboard **1006** receives ACK, which indicate the success of previous HID transmission. The keyboard **1006** also receives Chroma data to configure the lighting effect of the keyboard **1006**.

(62) In some embodiments, one single RF channel may be used instead of a subset of sub-carrier frequency. In some embodiments, a single wireless protocol is deployed. In some embodiments, the dongle (receiver) may control the time slots, and both mouse and keyboard may be synchronized to the dongle. The packet transmission is time slot base instead of contention base.

(63) In some embodiments, the synchronization is controlled by the dongle. The dongle may transmit a beacon on each time slot upon power up (before the respective HID device is connected). The HID device (e.g., mouse, keyboard) may listen for beacon upon power up. The address of each beacon may be assigned according to the time slot assignment. In another words, mouse may only receive mouse beacon, whereas keyboard may only receive keyboard beacon. So the beacons for different HID devices may not mix up and compete for the same timeslot. Once the mouse (or keyboard) receives the beacon, it may perform the connection process with the dongle using the beacon as the reference for the time slot synchronization.

(64) In some embodiments, the duration of a frame may be 500 μ s (microsecond). Each frame may include at least three time slots. The time slot for mouse and keyboard may be 150 μ s each. And the time slot for Chroma data may be 200 μ s. Frame duration of 500 μ s may provide a frame rate of 2000 Hz. The minimum frame rate for a HID device is 1000 Hz. So there may be 100% redundancy for error recovery. The 150 μ s time slot for mouse and keyboard may be determined by the packet size of each transmission. Each mouse or keyboard packet may include 8 bytes of data.

Thus, a 150 μ s time slot may be sufficient to complete the data transmission. And the remaining time slot may be assigned to Chroma data.

(65) FIG. **11** is a flowchart **1100** of a method of wireless communication. In some embodiments, the method may operate a dongle in the data transmission system described above with reference to FIG. **2**. The method may be performed by an apparatus (e.g., apparatus **1202/1202'** described below with reference to FIG. **12** or **13**). The apparatus may be a dongle. At **1102**, the apparatus may receive, via a radio frequency channel, a first data from a first human interface device at a first time slot of a frame.

(66) At **1104**, the apparatus may transmit the first data to a computing device coupled to the apparatus. In some embodiments, the apparatus may be coupled to the computing device through a USB port of the computing device.

(67) At **1106**, the apparatus may receive, via the radio frequency channel, a second data from a second human interface device at a second time slot of the frame. In some embodiments, the first human interface device and the second human interface device may be a wireless mouse and a wireless keyboard.

(68) In some embodiments, both the first human interface device and the second human interface device may be synchronized to the apparatus. In some embodiments, the apparatus may transmit, at the first time slot, a first beacon to synchronize with the first human interface device. The apparatus may further transmit, at the second time slot, a second beacon to synchronize with the second human interface device.

(69) At **1108**, the apparatus may transmit the second data to the computing device. In some embodiments, the first data may include movement and button status of the wireless mouse. In some embodiments, the second data may include keys pressed status of the wireless keyboard.

(70) At **1110**, the apparatus may transmit, via the radio frequency channel at a third time slot of the frame, at least one of a first acknowledgment to the first data, a first lighting effect configuration data for the first human interface device, a second acknowledgment to the second data, or a second lighting effect configuration data for the second human interface device. In some embodiments, the first lighting effect configuration data may be used to configure lighting effect of the first human interface device. In some embodiments, the second lighting effect configuration data may be used to configure lighting effect of the second human interface device.

(71) FIG. **12** is a conceptual data flow diagram **1200** illustrating the data flow between different means/components in an exemplary apparatus **1202**. The apparatus **1202** may be a dongle (e.g., the dongle **158** or **1002**). The apparatus **1202** may include a reception component **1204** that receives HID data from HID devices **1250** (e.g., a wireless mouse) and **1254** (e.g., a wireless keyboard). In one embodiment, the reception component **1204** may perform the operations described above with reference to **1102** or **1106** in FIG. **11**.

(72) The apparatus **1202** may include a transmission component **1210** that transmits ACK and Chroma data to HID device **1250** and/or **1254**. In one embodiment, the transmission component **1210** may perform the operations described above with reference to **1110** in FIG. **11**. The reception component **1204** and the transmission component **1210** may collaborate to coordinate the communication of the apparatus **1202**.

(73) The apparatus **1202** may include a data relay component **1206** that is configured to relay HID data from the reception component **1204** to a computing device **1252**. The relay component **1206** may be further configured to relay Chroma data from the computing device **1252** to the transmission component **1210**. In one embodiment, the data relay component **1206** may perform the operations described above with reference to **1104** or **1108** in FIG. **11**.

(74) The apparatus **1202** may include additional components that perform each of the blocks of the algorithm in the aforementioned flowcharts of FIGS. **3-5** and **11**. As such, each block in the aforementioned flowcharts of FIGS. **3-5** and **11** may be performed by a component and the apparatus may include one or more of those components. The components may be one or more

hardware components specifically configured to carry out the stated processes/algorithm, implemented by a processor configured to perform the stated processes/algorithm, stored within a computer-readable medium for implementation by a processor, or some combination thereof.

(75) FIG. 13 is a diagram 1300 illustrating an example of a hardware implementation for an apparatus 1202' employing a processing system 1314. In one embodiment, the apparatus 1202' may be the apparatus 1202 described above with reference to FIG. 12. The processing system 1314 may be implemented with a bus architecture, represented generally by the bus 1324. The bus 1324 may include any number of interconnecting buses and bridges depending on the specific application of the processing system 1314 and the overall design constraints. The bus 1324 links together various circuits including one or more processors and/or hardware components, represented by the processor 1304, the components 1204, 1206, 1210, and the computer-readable medium/memory 1306. The bus 1324 may also link various other circuits such as timing sources, peripherals, voltage regulators, and power management circuits, which are well known in the art, and therefore, will not be described any further.

(76) The processing system 1314 may be coupled to a transceiver 1310. The transceiver 1310 is coupled to one or more antennas 1320. The transceiver 1310 provides a means for communicating with various other apparatus over a transmission medium. The transceiver 1310 receives a signal from the one or more antennas 1320, extracts information from the received signal, and provides the extracted information to the processing system 1314, specifically the reception component 1204. In addition, the transceiver 1310 receives information from the processing system 1314, specifically the transmission component 1210, and based on the received information, generates a signal to be applied to the one or more antennas 1320.

(77) The processing system 1314 includes a processor 1304 coupled to a computer-readable medium/memory 1306. The processor 1304 is responsible for general processing, including the analyzation of data gathered by the apparatus itself through its own sensors and the execution of software stored on the computer-readable medium/memory 1306. The software, when executed by the processor 1304, causes the processing system 1314 to perform the various functions described supra for any particular apparatus. The computer-readable medium/memory 1306 may also be used for storing data that is manipulated by the processor 1304 when executing software. The processing system 1314 further includes at least one of the components 1204, 1206, 1210. The components may be software components running in the processor 1304, resident/stored in the computer readable medium/memory 1306, one or more hardware components coupled to the processor 1304, or some combination thereof.

(78) In the following, various aspects of this disclosure will be illustrated:

(79) Example 1 is a method or apparatus for wireless communication. The apparatus may receive, via a radio frequency channel, a first data from a first human interface device at a first time slot of a frame. The apparatus may transmit the first data to a computing device coupled to the apparatus. The apparatus may receive, via the radio frequency channel, a second data from a second human interface device at a second time slot of the frame. The apparatus may transmit the second data to the computing device. The apparatus may transmit, via the radio frequency channel at a third time slot of the frame, at least one of a first acknowledgment to the first data, a first lighting effect configuration data for the first human interface device, a second acknowledgment to the second data, or a second lighting effect configuration data for the second human interface device.

(80) In Example 2, the subject matter of Example 1 may optionally include that the first human interface device and the second human interface device may be a wireless mouse and a wireless keyboard.

(81) In Example 3, the subject matter of Example 2 may optionally include that the first data may include movement and button status of the wireless mouse.

(82) In Example 4, the subject matter of Example 2 may optionally include that the second data may include keys pressed status of the wireless keyboard.

(83) In Example 5, the subject matter of any one of Examples 1 to 4 may optionally include that the first lighting effect configuration data may be used to configure lighting effect of the first human interface device.

(84) In Example 6, the subject matter of any one of Examples 1 to 5 may optionally include that the second lighting effect configuration data may be used to configure lighting effect of the second human interface device.

(85) In Example 7, the subject matter of any one of Examples 1 to 6 may optionally include that both the first human interface device and the second human interface device may be synchronized to the apparatus.

(86) In Example 8, the subject matter of any one of Examples 1 to 7 may optionally include that the apparatus may transmit, at the first time slot, a first beacon to synchronize with the first human interface device; and may transmit, at the second time slot, a second beacon to synchronize with the second human interface device.

(87) It is understood that the specific order or hierarchy of blocks in the processes/flowcharts disclosed is an illustration of exemplary approaches. Based upon design preferences, it is understood that the specific order or hierarchy of blocks in the processes/flowcharts may be rearranged. Further, some blocks may be combined or omitted. The accompanying method claims present elements of the various blocks in a sample order, and are not meant to be limited to the specific order or hierarchy presented.

(88) The previous description is provided to enable any person skilled in the art to practice the various aspects described herein. Various modifications to these aspects will be readily apparent to those skilled in the art, and the generic principles defined herein may be applied to other aspects. Thus, the claims are not intended to be limited to the aspects shown herein, but is to be accorded the full scope consistent with the language claims, wherein reference to an element in the singular is not intended to mean “one and only one” unless specifically so stated, but rather “one or more.” The word “exemplary” is used herein to mean “serving as an example, instance, or illustration.” Any aspect described herein as “exemplary” is not necessarily to be construed as preferred or advantageous over other aspects. Unless specifically stated otherwise, the term “some” refers to one or more. Combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” include any combination of A, B, and/or C, and may include multiples of A, multiples of B, or multiples of C. Specifically, combinations such as “at least one of A, B, or C,” “one or more of A, B, or C,” “at least one of A, B, and C,” “one or more of A, B, and C,” and “A, B, C, or any combination thereof” may be A only, B only, C only, A and B, A and C, B and C, or A and B and C, where any such combinations may contain one or more member or members of A, B, or C. All structural and functional equivalents to the elements of the various aspects described throughout this disclosure that are known or later come to be known to those of ordinary skill in the art are expressly incorporated herein by reference and are intended to be encompassed by the claims. Moreover, nothing disclosed herein is intended to be dedicated to the public regardless of whether such disclosure is explicitly recited in the claims. The words “module,” “mechanism,” “element,” “device,” and the like may not be a substitute for the word “means.” As such, no claim element is to be construed as a means plus function unless the element is expressly recited using the phrase “means for.”

Claims

1. A method of wireless communication, the method comprising: receiving, at a receive mode of a device via a radio frequency channel, data from a human interface device at a respective time slot of a frame, wherein the frame comprises a sequence of time slots; relaying the received data to a computing device coupled to the device; and transmitting, at a transmit mode of the device via the

radio frequency channel at a last time slot of the frame, an acknowledgment and a lighting effect configuration data to the human interface device.

2. The method of claim 1, further comprising: setting an acknowledge flag for the human interface device by the device; and determining the lighting effect configuration data for the human interface device is ready, wherein the transmitting, at a transmit mode of the device via the radio frequency channel at a last time slot of the frame, an acknowledgment to the human interface device comprises transmitting the acknowledgment and the lighting effect configuration data for the human interface device to the human interface device.

3. The method of claim 2, further comprising: determining that lighting effect configuration data for the human interface device is not ready, wherein the transmitting, at a transmit mode of the device via the radio frequency channel at a last time slot of the frame, an acknowledgment to the human interface device comprises transmitting only the acknowledgment to the human interface device.

4. The method of claim 2, further comprising: receiving the acknowledgement and the lighting effect configuration data by the human interface device; and processing the acknowledgement and the lighting effect configuration data by the human interface device, wherein the lighting effect configuration data is used to configure lighting effect of the human interface device.

5. The method of claim 1, wherein durations of the sequence of time slots are allocated according to data bandwidth associated with the sequence of time slots, and a duration of the last time slot is greater than a duration of the respective time slot.

6. The method of claim 1, wherein the sequence of time slots comprises at least three time slots.

7. The method of claim 1, further comprising: transmitting a beacon at the respective time slot of the frame with an address associated with the human interface device.

8. The method of claim 1, further comprising: sending the data to the device by the human interface device; and setting a flag by the human interface device for the acknowledgement for the data.

9. An apparatus for wireless communication, the apparatus comprising: a memory; and at least one processor coupled to the memory and configured to: receive, at a receive mode of a device via a radio frequency channel, data from a human interface device at a respective time slot of a frame, wherein the frame comprises a sequence of time slots; relay the received data to a computing device coupled to the device; and transmit, at a transmit mode of the device via the radio frequency channel at a last time slot of the frame, an acknowledgment and a lighting effect configuration data to the human interface device.

10. The apparatus of claim 9, wherein the at least one processor is further configured to: set an acknowledge flag for the human interface device by the device; determine the lighting effect configuration data for the human interface device is ready; and transmit the acknowledgment and the lighting effect configuration data for the human interface device to the human interface device.

11. The apparatus of claim 10, wherein the at least one processor is further configured to: determine that lighting effect configuration data for the human interface device is not ready; and transmit only the acknowledgment to the human interface device.

12. The apparatus of claim 10, wherein the at least one processor is further configured to: receive the acknowledgement and the lighting effect configuration data by the human interface device; and process the acknowledgement and the lighting effect configuration data by the human interface device, wherein the lighting effect configuration data is used to configure lighting effect of the human interface device.

13. The apparatus of claim 9, wherein durations of the sequence of time slots are allocated according to data bandwidth associated with the sequence of time slots, and a duration of the last time slot is greater than a duration of the respective time slot.

14. The apparatus of claim 9, wherein the sequence of time slots comprises at least three time slots.

15. The apparatus of claim 9, wherein the at least one processor is further configured to: transmit a beacon at the respective time slot of the frame with an address associated with the human interface

device.

16. The apparatus of claim 9, wherein the at least one processor is further configured to: send the data to the device by the human interface device; and set a flag by the human interface device for the acknowledgement for the data.

17. A non-transitory computer-readable medium comprising computer executable instructions stored therein, which when executed by one or more processors, cause the one or more processors to perform: receiving, at a receive mode of a device via a radio frequency channel, data from a human interface device at a respective time slot of a frame, wherein the frame comprises a sequence of time slots; relaying the received data to a computing device coupled to the device; and transmitting, at a transmit mode of the device via the radio frequency channel at a last time slot of the frame, an acknowledgment and a lighting effect configuration data to the human interface device.

18. The non-transitory computer-readable medium of claim 17, wherein the instructions cause the one or more processors to further perform: setting an acknowledge flag for the human interface device by the device; and determining the lighting effect configuration data for the human interface device is ready, wherein the transmitting, at a transmit mode of the device via the radio frequency channel at a last time slot of the frame, an acknowledgment to the human interface device comprises transmitting the acknowledgment and the lighting effect configuration data for the human interface device to the human interface device.

19. The non-transitory computer-readable medium of claim 18, wherein the instructions cause the one or more processors to further perform: determining that lighting effect configuration data for the human interface device is not ready, wherein the transmitting, at a transmit mode of the device via the radio frequency channel at a last time slot of the frame, an acknowledgment to the human interface device comprises transmitting only the acknowledgment to the human interface device.

20. The non-transitory computer-readable medium of claim 18, wherein the instructions cause the one or more processors to further perform: receiving the acknowledgement and the lighting effect configuration data by the human interface device; and processing the acknowledgement and the lighting effect configuration data by the human interface device, wherein the lighting effect configuration data is used to configure lighting effect of the human interface device.
